

Loss Functions in Deep Learning

1 What is a Loss Function?

A **loss function** measures **how wrong** the model's predictions are.

It's like a **teacher giving feedback** — it tells the model:

“How far are you from the correct answer?”

The smaller the loss, the better the model is performing.

Example:

Suppose the model predicts:

- Predicted = 9
- Actual = 10

The loss could be:

$$\text{Loss} = |10 - 9| = 1$$

If prediction improves to 9.9, then:

$$\text{Loss} = |10 - 9.9| = 0.1$$

So, smaller loss = better performance.

2 Why Loss is Important

The **loss value** is used during **training** to:

1. Check how accurate the model is.
2. Guide the **gradient descent** algorithm to update weights.
3. Help the model gradually improve through many iterations.

Without a loss function, the model wouldn't know *what* to improve.

3 Loss Functions in Deep Learning

There are many types, depending on the problem:

(a) Regression Losses

Used when output is a number (e.g., predicting house price):

- **Mean Squared Error (MSE):**

$$L = \frac{1}{n} \sum (y_{true} - y_{pred})^2$$

→ Penalizes large errors more.

- **Mean Absolute Error (MAE):**

$$L = \frac{1}{n} \sum |y_{true} - y_{pred}|$$

→ Simpler, less sensitive to outliers.

(b) Classification Losses

Used when output is a class label (e.g., cat or dog):

- **Binary Cross Entropy (BCE):**
Used for 2 classes.
Measures how far predicted probabilities are from true labels.
- **Categorical Cross Entropy:**
Used for multiple classes (e.g., 10 digits in MNIST).
- **Hinge Loss:**
Common in SVMs, enforces a margin between classes.

(c) Advanced / Task-Specific Losses

- **Huber Loss:** Combines MAE and MSE (less sensitive to outliers).
- **KL Divergence:** Measures how one probability distribution differs from another.
- **Triplet Loss / Contrastive Loss:** Used in face recognition or similarity learning.

4 Loss Function vs. Cost Function

- **Loss Function:**
→ Error for **one** data point.
Example: loss for a single image or sample.
- **Cost Function:**
→ Average loss over **entire dataset (or batch)**.
Example: average loss across 100 training examples.

- So in practice:

$$\text{Cost} = \frac{1}{N} \sum_{i=1}^N \text{Loss}_i$$

5 Advantages & Disadvantages

Type	Advantages	Disadvantages
MSE	Smooth, easy to compute	Penalizes large errors heavily
MAE	Robust to outliers	Harder to optimize (not smooth)
Cross-Entropy	Works great for classification	Needs probabilistic output (sigmoid/softmax)
Hinge	Encourages correct margin	Not used for probabilistic outputs

6 How Loss Helps Training

During **backpropagation**:

- The loss function computes error.
- The network uses **gradient descent** to adjust weights.
- Step by step, the loss **gets smaller**, and predictions improve.

So loss function = the *compass* that points the model in the right direction.

In Summary:

- Loss = measure of model's mistake.
 - Smaller loss = better model.
 - Different losses for regression, classification, etc.
 - Cost = average loss over all samples.
 - It's central to how neural networks *learn and improve*.
-